Construction Sensitivity of a Shortest-Path Anisotropy Diagnostic in Geometric Graphs from Black-Hole Embeddings

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Abstract

Geometric-graph observables can depend on sampling, edge construction, and the choice of control. We examine this dependence for a shell-based shortest-path anisotropy statistic, $C_{\log}$, on graphs sampled from Schwarzschild/Flamm, Bardeen, and Hayward embedding surfaces. Canonical ten-seed Bardeen and Hayward parameter scans show organized but realization-dependent radial behavior. Paired audits at Schwarzschild/Flamm, Bardeen $g = 0.2$, and Hayward $\ell = 0.2$ compare five graph/control protocols using the same radial-angular sample and twelve common radial bins.

For Schwarzschild, mean absolute black-hole/control correlation gaps are 0.242 ± 0.161, 0.688 ± 0.442, 0.233 ± 0.199, 0.422 ± 0.323, and 0.681 ± 0.445 for the original, strict-radius, pure-KNN, mutual-KNN, and edge-matched-radius protocols (mean $\pm$ standard deviation over ten seeds). The Bardeen and Hayward anchors show the same qualitative construction dependence: original gaps are 0.255 ± 0.174 and 0.250 ± 0.165, whereas edge-matched-radius gaps are 0.676 ± 0.452 and 0.680 ± 0.443. Radius-style controls therefore do not erase the separation, and signed gaps vary across realizations and protocols.

The black-hole/control separation is neither construction invariant nor realization independent. $C_{\log}$ is reproducible when the complete sampling, construction, binning, seed, and null-model protocol is specified, but the evidence does not support interpreting it as a graph-independent curvature scalar. The contribution is a matched-control audit framework for attaching geometric claims to a complete diagnostic–construction–control protocol.

Keywords: geometric graphs; spatial networks; shortest-path multiplicity; graph construction; matched controls; black-hole embeddings

1 Introduction

Geometric graphs are widely used as finite representations of sampled metric spaces, spatial data, and physical or data-driven geometries [2, 3, 4]. However, a graph observable measured on such a representation is rarely a function of geometry alone. It can also depend on sampling density, boundary effects, edge rule, graph scale, and null model. Similar difficulties appear in thresholded and density-controlled network analyses, where changing the graph construction can change the interpretation of a measured network statistic [5, 6, 7]. For this reason, geometric interpretations of graph diagnostics require matched controls and explicit construction-sensitivity tests.

This issue is related, but not identical, to the problem of defining curvature on discrete structures. Classical discretizations such as Regge calculus encode curvature through a specific discretized geometric framework [8]. Graph and cell-complex notions such as Forman and Ollivier Ricci curvature provide alternative discrete analogues with different mathematical meanings [9, 10]. The present work takes a complementary approach. We do not attempt to define a new discrete curvature tensor or scalar. Instead, we ask when a shortest-path graph diagnostic displays geometric organization and whether that organization survives stricter matched-control tests.

We use embedded spatial slices of black-hole geometries as a controlled setting for this question. These surfaces provide smooth radial benchmarks rather than new astrophysical predictions. The benchmark families are the Schwarzschild/Flamm embedding and two regular black-hole models, Bardeen and Hayward [11, 12, 13, 14]. Under the original fixed construction, representative graphs from all three families show closely aligned radial $C_{\log}$ profiles. This observation motivates, but does not by itself establish, a construction-independent geometric signal.

The diagnostic $C_{\log}$ measures the dispersion of logarithmic shortest-path multiplicities over a finite graph-distance shell. An earlier single-realization construction sweep suggested that the black-hole/control gap was large under the original matched-flat KNN control and almost vanished under radius-style controls. The purpose of the present revision is to test that interpretation with a paired multi-seed audit and a single canonical correlation pipeline. The audit shows that the earlier collapse was not representative: strict-radius and edge-matched-radius controls instead produce the largest mean absolute gaps, while realization-to-realization variability is substantial.

The scientific message is therefore not that one construction is uniquely correct. The separation depends jointly on the diagnostic, graph construction, and control. We distinguish three logically different questions: (i) reproducibility within a fixed protocol, (ii) sensitivity across protocols, and (iii) invariance under graph construction. The ten-seed experiment quantifies the first two and does not support the third.

We report canonical ten-seed fixed-protocol scans for Bardeen and Hayward and representative five-protocol audits for both families. These data separate reproducibility within a protocol from sensitivity across constructions; they do not establish graph-independent curvature invariance.

An earlier version of this work reported fixed-protocol and single-realization results [1]. The present manuscript replaces that construction analysis with paired ten-seed audits, adds canonical Bardeen and Hayward calculations, and removes quantitative claims that did not survive the multi-seed validation.

2 Graph benchmarks and matched controls

2.1 Embedding geometries

We consider geometric graphs sampled from embedded spatial slices of static black-hole benchmark geometries. For a static, spherically symmetric metric written as

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \tag{1}$$

the equatorial spatial slice has line element

$$dl^2 = f(r)^{-1}dr^2 + r^2 d\phi^2. \tag{2}$$

Embedding this slice as a surface of revolution in Euclidean three-space gives

$$dl^2 = \left(1 + \left(\frac{dz}{dr}\right)^2\right) dr^2 + r^2 d\phi^2, \tag{3}$$

and therefore

$$\frac{dz}{dr} = \sqrt{\frac{1}{f(r)} - 1}. \tag{4}$$

The Schwarzschild case uses

$$f_{\mathrm{Schw}}(r) = 1 - \frac{2M}{r}. \tag{5}$$

For the Bardeen regular black-hole family, we use

$$f_{\mathrm{B}}(r) = 1 - \frac{2Mr^2}{(r^2 + g^2)^{3/2}}, \tag{6}$$

and for the Hayward family,

$$f_{\mathrm{H}}(r) = 1 - \frac{2Mr^2}{r^3 + 2M\ell^2}. \tag{7}$$

The scans use $M = 1/2$, $g \in \{0, 0.1, 0.2, 0.3\}$, and $\ell \in \{0, 0.1, 0.2, 0.3\}$. The five-protocol anchors use $g = 0.2$ and $\ell = 0.2$.

2.2 Matched-flat controls

For each black-hole graph, matched-flat controls reuse the same radial and angular samples. The flat control removes the embedded height,

$$(r, \phi, z(r)) \longrightarrow (r, \phi, 0), \tag{8}$$

while preserving the radial-angular sampling. The control therefore asks how much of the diagnostic organization remains after replacing the curved embedding by a flat geometry with the same sampling information.

The graph-construction rule is part of the protocol. A diagnostic value is not interpreted independently of the edge rule and null model; every comparison is treated as a diagnostic–construction–control triple.

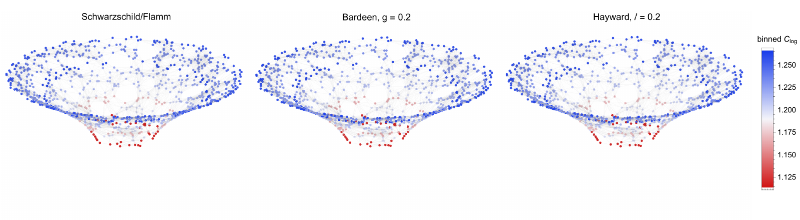

Figure 1: Representative Schwarzschild/Flamm, Bardeen ($g = 0.2$), and Hayward ($\ell = 0.2$) embedding graphs, colored by radially binned mean $C_{\log}$ on a common scale. These fixed-protocol visualizations motivate the broader family program; they are not a substitute for the paired multi-protocol audit reported below.

Alt text: Three colored three-dimensional geometric graphs show Schwarzschild/Flamm, Bardeen, and Hayward embedding surfaces. A shared color scale reveals comparable radial organization across the three graph families.

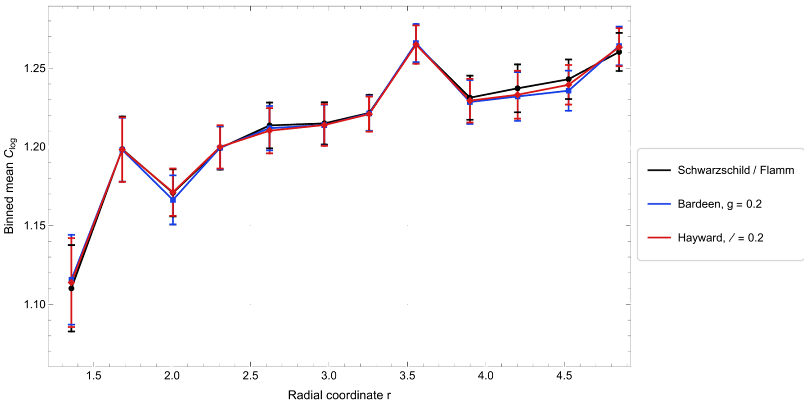

Figure 2: Representative binned radial $C_{\log}$ profiles for the three benchmark families under the original calibrated construction. The close alignment shows that the motivating radial organization is not visually unique to Schwarzschild; the canonical scans and anchor audits below provide the quantitative family comparison.

Alt text: Binned mean $C_{\log}$ is plotted against radial coordinate for Schwarzschild/Flamm, Bardeen $g = 0.2$, and Hayward $\ell = 0.2$. The three profiles and error bars closely track one another.

3 Shortest-path anisotropy diagnostic

Let $G = (V, E)$ be a finite graph and let $d_G(i, j)$ denote graph distance. For a reference vertex i and graph-shell radius r_g, define

$$S_{r_g}(i) = \{j \in V : d_G(i, j) = r_g\}. \tag{9}$$

Let m_{ij} be the number of shortest paths between i and j. The statistic $C_{\log}$ is the cubic mean deviation of the logarithmic multiplicities over the shell,

$$C_{\log}(i; r_g) = \left(\frac{1}{|S_{r_g}(i)|} \sum_{j \in S_{r_g}(i)} \left| \log m_{ij} - \langle \log m \rangle_{S_{r_g}(i)} \right|^3 \right)^{1/3}. \tag{10}$$

This quantity probes anisotropy in shortest-path multiplicity patterns rather than degree or edge density alone. It is a finite-shell graph observable, not a continuum scalar by definition.

3.1 Canonical numerical protocol

All audited graphs use $M = 1/2$, $N = 1000$, construction parameter $k = 16$, graph-shell radius $r_g = 3$, twelve common radial bins, and seeds 1–10. One radial-angular sample is generated per family, parameter, and seed. Its flat control reuses that sample; at each anchor the same sample is also reused by all five graph/control protocols. The calibrated radius convention retains the factor 1.15 used in the original implementation. Pearson radial correlations are primary; Spearman correlations and Pearson correlation of the black-hole common-bin $C_{\log}$ profile with $\log K$ on those same bins are secondary.

Vertex-level $C_{\log}$ values are evaluated first. The black-hole and flat values are then binned on the same overlapping radial domain. Pearson correlations are computed between the common-bin mean radius and common-bin mean $C_{\log}$ profile,

$$\rho_{\mathrm{BH}} = \mathrm{Corr}(r, C_{\log})_{\mathrm{BH}}\,, \qquad \rho_{\mathrm{flat}} = \mathrm{Corr}(r, C_{\log})_{\mathrm{flat}}\,, \tag{11}$$

with

$$\Delta\rho_r = \rho_{\mathrm{BH}} - \rho_{\mathrm{flat}}. \tag{12}$$

We report signed $\Delta\rho_r$ to retain direction and $|\Delta\rho_r|$ to summarize separation magnitude. No sign reversal to $-C_{\log}$ is used.

4 Paired construction-sensitivity audit

The audit tests whether the Schwarzschild/Flamm black-hole/control separation is stable under reasonable changes in the edge rule and flat-control construction. The five protocols are summarized in Table 1.

Table 1: Graph-construction and matched-control protocols used in the paired ten-seed audit.

Protocol	Black-hole graph	Flat control
Original matched-flat KNN	Calibrated radius-threshold graph	Matched-flat KNN graph
Strict radius	Calibrated radius-threshold graph	Radius-threshold graph from the flat sample
Pure KNN	Symmetrized KNN graph	Symmetrized KNN graph
Mutual KNN	Mutual-KNN graph	Mutual-KNN graph
Edge-matched radius	Calibrated radius-threshold graph	Radius graph matched to the black-hole edge count

All 50 runs passed the preregistered quality checks: ten rows per protocol, five rows per seed, twelve common bins, no missing correlations, maximum coordinate mismatch 1.33×10^{-15}, connected black-hole and flat graphs, unit largest-component and valid-$C_{\log}$ fractions, valid mutual-KNN subset checks, and exact or tie-explained edge matching. No seed was removed.

4.1 Protocol-level results

Table 2 reports the signed and absolute gaps. The result is not a monotone correction toward zero. Strict-radius and edge-matched-radius controls give the largest mean absolute separations, while the original and pure-KNN protocols give smaller averages. Strict radius and edge-matched radius are almost identical in paired mean absolute gap even though their control radii are selected differently.

Table 2: Ten-seed Schwarzschild/Flamm construction audit. Values are mean $\pm$ sample standard deviation. The 95% interval is a two-sided Student-t interval for the mean of $|\Delta\rho_r|$ with nine degrees of freedom.

Protocol	Signed $\Delta\rho_r$	$\lvert\Delta\rho_r\rvert$	95% CI for mean $\lvert\Delta\rho_r\rvert$
Original matched-flat KNN	0.160 ± 0.250	0.242 ± 0.161	$[0.126, 0.357]$
Strict radius	0.688 ± 0.442	0.688 ± 0.442	$[0.372, 1.005]$
Pure KNN	-0.189 ± 0.245	0.233 ± 0.199	$[0.091, 0.375]$
Mutual KNN	0.392 ± 0.362	0.422 ± 0.323	$[0.192, 0.653]$
Edge-matched radius	0.681 ± 0.445	0.681 ± 0.445	$[0.363, 1.000]$

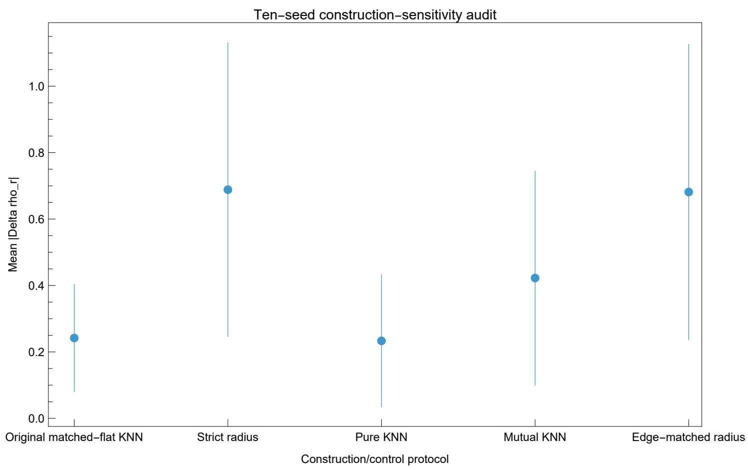

Figure 3: Mean absolute radial-correlation gap across the ten paired seeds. Error bars show sample standard deviation. The broad error bars are part of the result: protocol sensitivity and realization variability are both substantial.

Alt text: Mean absolute black-hole/control correlation gap with standard-deviation error bars for five graph protocols. The two radius-based controls have the largest means and broad variation.

The separation is not explained by a single edge-density comparison. Table 3 shows that strict radius and edge-matched radius give nearly the same gap even though the former has a small seed-dependent edge-count mismatch and the latter is exactly matched. Conversely, original and pure KNN have similar mean absolute gaps despite very different black-hole edge rules.

Table 3: Mean edge counts across ten seeds. The original protocol has a large black-hole/flat density mismatch, whereas the other symmetric protocols are approximately or exactly edge matched.

Protocol	Mean BH edges	Mean flat edges	Mean BH–flat difference
Original matched-flat KNN	10171.2	9044.4	1126.8
Strict radius	10171.2	10156.4	14.8
Pure KNN	9046.5	9044.4	2.1
Mutual KNN	6953.5	6955.6	-2.1
Edge-matched radius	10171.2	10171.2	0.0

4.2 Realization dependence and paired contrasts

Figures 4 and 5 display the same ten point samples across all protocols. The ordering is not identical from seed to seed. Original, pure-KNN, and mutual-KNN gaps change sign in some realizations; strict-radius and edge-matched-radius gaps are positive in all ten runs. The original protocol contains one conventional $1.5 \times$ IQR outlier at seed 10, $\Delta\rho_r = 0.6102$. Its graph, pairing, bins, connectivity, and estimator-validity checks all pass, so it is retained.

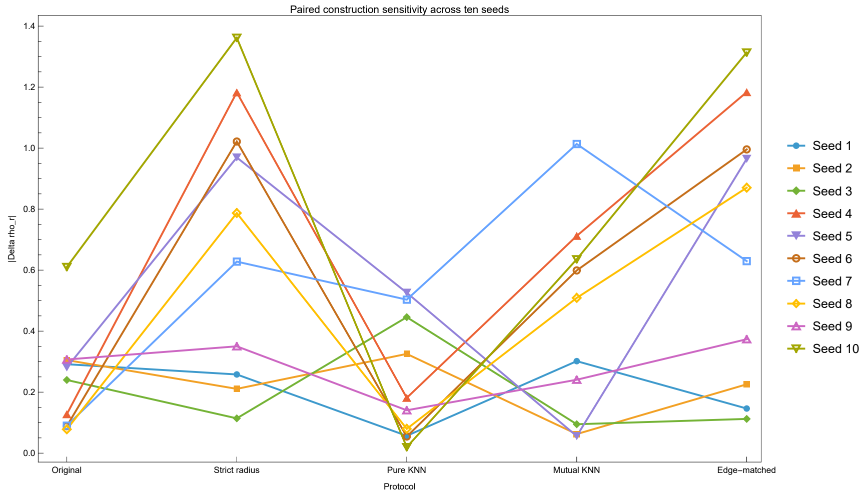

Figure 4: Paired $|\Delta\rho_r|$ values for all ten seeds. Lines join the same radial-angular sample across graph/control protocols.

Alt text: Ten colored seed trajectories show absolute correlation gap across five categorical protocols. The ordering changes by seed, with frequent peaks at the radius-based controls.

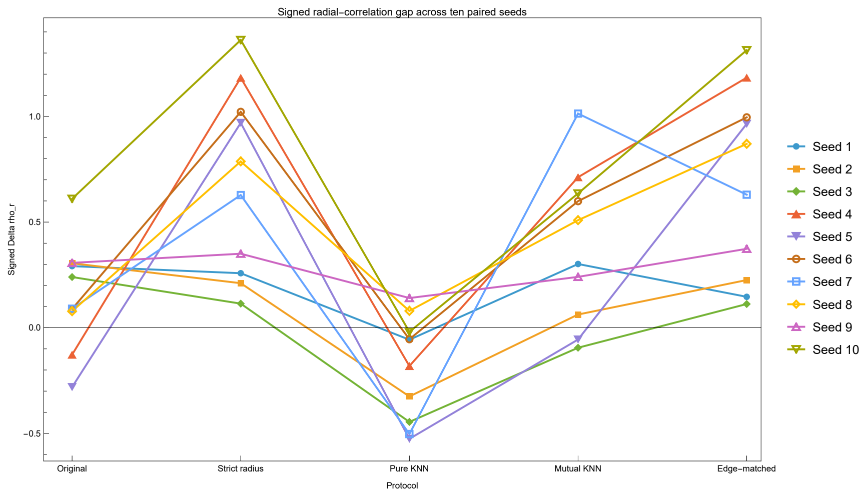

Figure 5: Signed $\Delta\rho_r$ for the same paired realizations. The sign changes show why signed and absolute gaps must be reported separately.

Alt text: Ten paired seed trajectories show signed correlation gap across five protocols. Several trajectories cross zero, especially for original, pure-KNN, and mutual-KNN graphs.

Selected paired contrasts are given in Table 4. The strict-radius and edge-matched-radius means exceed the original and pure-KNN means in this sample, while strict radius and edge-matched radius are mutually indistinguishable. These intervals quantify the paired experiment; they do not establish a universal ranking of graph constructions.

Table 4: Selected paired contrasts in $|\Delta\rho_r|$. The contrast is protocol A minus protocol B; intervals are two-sided Student-t intervals over ten paired seeds.

Protocol A	Protocol B	Mean paired difference	95% CI
Strict radius	Original matched-flat KNN	0.446	$[0.122, 0.771]$
Strict radius	Pure KNN	0.455	$[0.071, 0.839]$
Edge-matched radius	Original matched-flat KNN	0.440	$[0.108, 0.771]$
Edge-matched radius	Pure KNN	0.448	$[0.066, 0.831]$
Strict radius	Edge-matched radius	0.007	$[-0.029, 0.043]$
Original matched-flat KNN	Pure KNN	0.009	$[-0.192, 0.209]$

5 Canonical Bardeen and Hayward results

The fixed-protocol scans quantify the family context that was previously only illustrative. Table 5 reports mean $\pm$ sample standard deviation over ten paired seeds. The flat-control values repeat across parameter values because the same seed convention and radial-angular sampling law are used. Increasing either regularization parameter changes the black-hole profile gradually compared with the large realization spread. The black-hole common-bin profile is negatively related to $\log K$ on average; for example, the mean Pearson correlations at parameter 0.2 are -0.674 ± 0.258 (Bardeen) and -0.681 ± 0.245 (Hayward). These are associations on the same twelve bins, not evidence that $C_{\log}$ is a curvature scalar.

Table 5: Canonical original-protocol family scans (mean $\pm$ SD, ten seeds).

Family	Parameter	ρ_{BH}	ρ_{flat}	$\Delta\rho_r$	$\|\Delta\rho_r\|$
Bardeen	0.0	0.592 ± 0.266	0.432 ± 0.201	0.160 ± 0.250	0.242 ± 0.161
	0.1	0.587 ± 0.275	0.432 ± 0.201	0.155 ± 0.258	0.248 ± 0.159
	0.2	0.581 ± 0.293	0.432 ± 0.201	0.149 ± 0.279	0.255 ± 0.174
	0.3	0.563 ± 0.331	0.432 ± 0.201	0.131 ± 0.316	0.272 ± 0.192
Hayward	0.0	0.592 ± 0.266	0.432 ± 0.201	0.160 ± 0.250	0.242 ± 0.161
	0.1	0.590 ± 0.271	0.432 ± 0.201	0.158 ± 0.254	0.246 ± 0.159
	0.2	0.588 ± 0.278	0.432 ± 0.201	0.156 ± 0.263	0.250 ± 0.165
	0.3	0.579 ± 0.292	0.432 ± 0.201	0.147 ± 0.276	0.254 ± 0.169

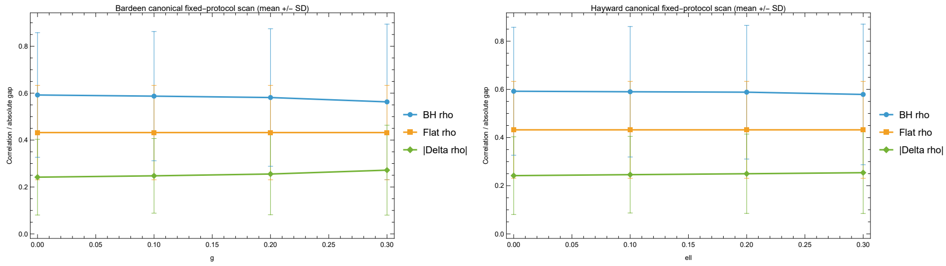

Figure 6: Canonical Bardeen (left) and Hayward (right) parameter scans. Error bars show sample SD across paired seeds.

Alt text: Side-by-side Bardeen g and Hayward ℓ parameter scans compare black-hole radial correlation, flat-control correlation, and absolute gap. Parameter shifts are modest relative to the standard-deviation error bars.

The anchor audits show that the Schwarzschild construction sensitivity generalizes qualitatively. For Bardeen, mean $|\Delta\rho_r|$ is 0.255 ± 0.174, 0.678 ± 0.435, 0.211 ± 0.170, 0.422 ± 0.322, and 0.676 ± 0.452 in protocol order. For Hayward the corresponding values are 0.250 ± 0.165, 0.684 ± 0.441, 0.216 ± 0.200, 0.413 ± 0.337, and 0.680 ± 0.443. Thus radius-style controls again give the largest average separation, but broad paired variation precludes a universal protocol ranking.

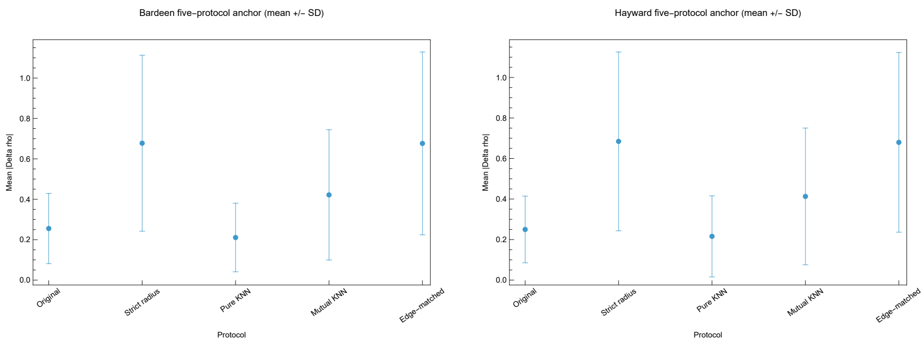

Figure 7: Five-protocol anchors for Bardeen $g = 0.2$ (left) and Hayward $\ell = 0.2$ (right), shown as mean $|\Delta\rho_r| \pm$ SD. Seed-level signed and absolute plots are retained with the reproducibility outputs.

Alt text: Side-by-side five-protocol error-bar plots for Bardeen $g = 0.2$ and Hayward $\ell = 0.2$. Both show smaller original and pure-KNN gaps and larger radius-control gaps with broad uncertainty.

The zero-parameter rows reproduce Schwarzschild row by row, not merely in their means: coordinate hashes, common-profile hashes, and every exported numeric output agree for each seed to tolerance 10^{-12} (maximum observed difference zero). Unsupported legacy claims of a common 0.78 gap and a radius-control collapse to 10^{-3} are therefore removed rather than mixed with the canonical pipeline.

6 Discussion

The paired audits change, but do not erase, the original result. $C_{\log}$ displays organized radial behavior across the three black-hole embedding families. The question is whether the difference from a flat control is stable when the graph representation changes. It is not. Both the magnitude and the sign of the gap depend on the full protocol and on the random realization.

The most important correction is that radius-style controls do not absorb the signal in any canonical anchor experiment. Strict-radius and edge-matched-radius controls instead produce the largest mean absolute gaps. Their near equality is especially informative: exact edge-count matching does not remove the effect, and a small residual edge-count difference in the strict-radius protocol does not explain it. At the same time, the broad seed-to-seed spread prevents a universal protocol ranking.

The result supports a more precise interpretation of $C_{\log}$. It is a statistic of finite-shell shortest-path multiplicity organization. Shortest paths are jointly determined by the sampled surface and by the imposed graph representation. The appropriate object of interpretation is therefore

$$\text{diagnostic} + \text{construction} + \text{control}, \tag{13}$$

not the diagnostic alone.

This distinction matters beyond black-hole embeddings. Geometric-graph studies should report the sampling prescription, graph scale, edge rule, shell radius, boundary handling, binning convention, seed distribution, and null-model construction. At minimum, a geometric claim should be tested against a sampling-matched control, a construction-matched control, and an edge-density or edge-count-matched control.

6.1 Scope and limitations

The present experiment is a finite-graph benchmark, not a continuum curvature reconstruction. It uses one graph size, one shell radius, one bin count, and ten paired realizations. The confidence intervals are broad, and the analysis does not establish continuum convergence. The graphs are built from Euclidean embeddings of spatial slices; relationships to intrinsic two-dimensional curvature, four-dimensional spacetime invariants, or non-embeddable data require separate analysis.

The Bardeen and Hayward construction audits cover one representative parameter each rather than the complete protocol-by-parameter matrix. The paper therefore does not claim cross-family construction invariance or a universal ordering of protocols. Multi-resolution runs, shell-radius sweeps, alternative binning, additional parameter anchors, and canonical audits of other diagnostics remain useful extensions.

6.2 Reproducibility

The new calculation contains exactly 80 family-scan rows and 100 anchor-audit rows, with no missing family, parameter, seed, or protocol. All primary and secondary correlations are numeric, every run contains twelve common radial bins, and the maximum radial-angular mismatch is 1.33×10^{-15}. Complete vertex sets, connectivity, valid-$C_{\log}$ fractions, mutual-KNN subset checks, and exact or tie-explained edge matching are recorded. The 100 checkpoint units required 27701.3 seconds (7.69 h) of compute time, ranging from 116.5 to 1099.9 seconds per checkpoint. The preceding Schwarzschild audit required 3636.84 seconds.

Code, scripts, raw data, quality-control tables, checkpoints, validation files, and manuscript figures are available in the project repository. A DOI-backed archival snapshot should accompany journal submission.

7 Conclusion

We studied the shortest-path anisotropy statistic $C_{\log}$ on geometric graphs sampled from embedded black-hole spatial slices. Canonical ten-seed Bardeen and Hayward scans show organized radial behavior under one fixed protocol, with family-parameter and realization dependence. Paired five-protocol audits at Schwarzschild and representative Bardeen and Hayward anchors test how the black-hole/control separation changes with graph construction.

The audit does not confirm the earlier claim that radius-style controls erase the gap. Strict-radius and edge-matched-radius constructions instead produce the largest mean absolute separations, whereas the original and pure-KNN protocols produce smaller averages. Signed gaps vary across realizations and reverse under some protocols. The separation is therefore neither graph-construction invariant nor realization independent.

The main conclusion is methodological. $C_{\log}$ is a reproducible finite-shell shortest-path statistic under fully specified conditions, but it is not a graph-independent curvature scalar. Apparent geometric signals in spatial graph diagnostics should be interpreted as properties of complete diagnostic–construction–control protocols. The anchor results show that strong construction dependence is shared across the three tested families, while limited parameter, resolution, and shell coverage prevents a graph-independent curvature interpretation.

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Conflict of interest

The author declares no conflict of interest.

Data and code availability

Scripts, raw data, validation outputs, and figure-generation materials are available from https://github.com/Uxuee/PathAnisotropyCurvature.
ZENODO DOI REQUIRED BEFORE SUBMISSION

AI-assisted tools

OpenAI ChatGPT and Codex were used to assist with language editing, code organization, reproducibility checks, and preparation of the manuscript materials. The author reviewed and verified the text, code, analyses, figures, and numerical results and takes full responsibility for the contents of the work.

References

[1] A. U. Palomino Ylla, Construction sensitivity of shortest-path anisotropy diagnostics in geometric graphs from black-hole embeddings, *arXiv preprint* arXiv:2606.27248 (2026).

[2] M. Penrose, *Random Geometric Graphs*. Oxford University Press, Oxford, 2003.

[3] J. Dall and M. Christensen, Random geometric graphs, *Physical Review E* **66**, 016121 (2002).

[4] M. E. J. Newman, *Networks: An Introduction.* Oxford University Press, Oxford, 2010.

[5] B. C. M. van Wijk, C. J. Stam, and A. Daffertshofer, Comparing brain networks of different size and connectivity density using graph theory, *PLoS ONE* **5**, e13701 (2010).

[6] N. Langer, A. Pedroni, and L. Jäncke, The problem of thresholding in small-world network analysis, *PLoS ONE* **8**, e53199 (2013).

[7] C. E. Ginestet, T. E. Nichols, E. T. Bullmore, and A. Simmons, Brain network analysis: separating cost from topology using cost-integration, *PLoS ONE* **6**, e21570 (2011).

[8] T. Regge, General relativity without coordinates, *Nuovo Cimento* **19**, 558–571 (1961).

[9] R. Forman, Bochner's method for cell complexes and combinatorial Ricci curvature, *Discrete & Computational Geometry* **29**, 323–374 (2003).

[10] Y. Ollivier, Ricci curvature of Markov chains on metric spaces, *Journal of Functional Analysis* **256**, 810–864 (2009).

[11] K. Schwarzschild, Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie, *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften* 189–196 (1916).

[12] L. Flamm, Beiträge zur Einsteinschen Gravitationstheorie, *Physikalische Zeitschrift* **17**, 448–454 (1916).

[13] J. M. Bardeen, Non-singular general-relativistic gravitational collapse, in *Proceedings of the International Conference GR5*, Tbilisi, USSR, p. 174 (1968).

[14] S. A. Hayward, Formation and evaporation of nonsingular black holes, *Physical Review Letters* **96**, 031103 (2006).